

Benjamin G. Pinoli

pinoliben@gmail.com | 503-883-1762 | linkedin.com/in/ben-pinoli | github.com/benpinoli

EDUCATION

B.S. in Applied Computational Mathematics, ML Emphasis, Minors in Physics and Business

Apr 2028

Brigham Young University

Provo, UT

- Cumulative GPA: 3.97, Major GPA: 4.0, 4-Year Half Tuition Scholarship from BYU, 2X Dean's List
- BYU AIA (Competitive ML Team), Computational Finance IDEa Labs, AI in Business Society, Quantum Information Association
- Selected Coursework: Multivariable Calculus, ODEs, Computational Linear Algebra (Python), Data Structures, Waves and Optics, Error and Recursion (Python, C++), Econometrics, Advanced Machine Learning, Stochastic Models, Theory of Analysis, Numerical Optimization, Asset Pricing Theory, Gaussian Processes Regression, Deep Metric Learning, Kernel Methods, Clustering, RVM

TECHNICAL SKILLS

Languages: Python, C++, SQL

Libraries/Frameworks: PyTorch, Pandas, Polars, PySpark, Scikit-Learn, MLFlow

Systems/Infra: AWS (EC2, Amplify), Supabase, Git, Docker, PostgreSQL

WORK EXPERIENCE

Machine Learning and Data Engineering Intern

May 2026-Present

BYU-Pathway Worldwide

Salt Lake City, UT

- Achieved .83 AUC and 85% accuracy in binary prediction models with highly irregular data through optimized feature engineering
- Engineered ML-oriented silver table EAV schema integrating 10+ sources, robustly architected for future variables
- Created novel ranking and GLM interpretability methods to 1) define fastest-path improvement for students, 2) quantify suggested feature changes (career development feedback) per-student, 3) dynamically update improvement vector per-student with new data
- Modeling 80,000 students worldwide; projected to process 1M+ by 2030 as BYU-Pathway's TalentStorehouse scales globally

Co-Founder, Systems Engineer

May 2026-Present

Bear Automation

Provo, UT

- Spearhead technical sales for custom DME company and law firm software; operate as on-site FDE to map businesses' ontologies
- Pilot testing of automated workflow indicates a potential savings of 5-15 hours per employee weekly upon full company rollout
- Expected annual PnL increase of 1MM+ USD upon company-wide adoption, transforming efficiency and operations

Quantitative Researcher

Jan 2026-Apr 2026

BYU Silver Fund

Provo, UT

- Conducted signal research for 5M USD portfolio through specialized package sf-quant under direction of Xantium PM Dr. Bates
- Constructed novel Kullback-Leibler divergence, MI techniques in factor model residuals against S&P volatility
- Applied EWMA rolling PCA to idiosyncratic returns in competitive market subsets to train predictive DLM
- Created weak supervision model, 10.7% annual alpha out-of-sample, placed third; judges from Two Sigma, Bridgewater, Xantium

PERSONAL PROJECTS

Competitive Market Subsets DLM (github.com/benpinoli/market_subset_DLM) | Python

Mar 2026

- Obtained rolling principal component of idiosyncratic returns in Russell 3000, binned by NAICS six-digit codes
- Leveraged sf-quant package and distributed lag model to explore competition dynamics across daily, weekly, monthly idiosyncratic returns

Nonlinear Idiosyncratic Returns (github.com/benpinoli/Mutual-Information-across-stock-returns) | Python

Feb 2026

- Utilized sf-quant package on Barra factor model residuals plotted on VIX, leveraged information theory to identify trading opportunities
- Theorized novel zero-factor-exposure trading strategy, discovered signal's non-orthogonality with classic idiosyncratic reversion technique

PatientOS (github.com/benpinoli/PatientOS) | TypeScript, PLpgSQL

Jun 2026

- Delivered custom software via AWS Amplify, obeying company ontology and hierarchy
- Configured EC2-hosted Supabase instance to architect scalable EAV-organized patient database
- Integrated Gemini API for autonomous form-writing for robust template-based forms

Roofing Area Estimator (github.com/benpinoli/roof-snap) | Python

May 2026

- Leveraged Gemini API, Google Solar API and graph theory to create website for obtaining 80% accurate roof area estimates
- UI includes cost calculations for various building materials; estimates hips, valleys, ridges, and eaves for every address on Google Solar

Fish & Gravitation Physics (github.com/benpinoli/Fish-and-Gravitation-Physics) | Python

Apr 2026

- Coded gravitation and fish simulation; fully tensorized calculations for various forces between generalized n-body situations

Other Projects: ChineseCheat (payhip.com/b/ygkre), "Shane Street" Informed Trading (LinkedIn), Podcast App with Bear Automation

SERVICE AND OTHER

Volunteer Service Representative

Jun 2023-Jun 2025

The Church of Jesus Christ of Latter-day Saints

Taichung, Taiwan

- Managed digital marketing strategy, streamlining Mandarin Chinese media production, and quintupled annualized leads supported by .5M USD over 7.5 months in southern Taiwan; trained 70 leaders over ~5,000 volunteers across Asia
- Developed on-field technology methods to triple RLA (Rapid Language Acquisition) speed and transform long term retention

Languages: Fluent in English and Mandarin | **Awards & Scores:** SAT: 1550, 2025 BYU Battle of the Quants Finalist, Eagle Scout